NATURAL SCIENCES TRIPOS Part II
PHYSICS (4) PHYSICAL SCIENCES: HALF SUBJECT PHYSICS (4)

ADVANCED AND QUANTUM OPTICS

Candidates offering this paper should attempt all six questions (four from Section A and two from Section B).
The approximate number of marks allocated to each question or part of a question is indicated in the right margin. This paper contains 4 sides, including this coversheet, and is accompanied by a handbook giving values of constants and containing mathematical formulae which you may quote without proof.

STATIONERY REQUIREMENTS
2 × 8-page answer booklet
Rough workpad
Yellow master coversheet

SPECIAL REQUIREMENTS
Mathematical Formulae Handbook
Approved calculator allowed

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the Invigilator.

SECTION A

Attempt all four questions in this Section. Answers should be concise and relevant formulae may be assumed without proof.

1. Define the numerical aperture (NA) and the Abbe resolution limit for a microscope. Explain briefly how the use of immersion oil can improve the spatial resolution compared to imaging in air.

2. Explain the concept of population inversion in a laser medium. Show mathematically why a medium in thermal equilibrium attenuates rather than amplifies an incident optical wave.

3. Define the temporal coherence time and coherence length for random light. Given a Light Emitting Diode with a central wavelength $\lambda_0 = 1\,\mu$m and a spectral width $\Delta\lambda_0 = 50$ nm, calculate its coherence length.

4. State the general phase-matching condition for three-wave mixing. Explain qualitatively how birefringence in a uniaxial crystal can be used to satisfy this condition for collinear Second-Harmonic Generation (SHG).

SECTION B

Attempt both questions from this Section.

5. This question considers the pumping and rate equations of a laser amplifier.

   (a) Describe the underlying principle of Light Amplification by Stimulated Emission of Radiation (LASER).

   (b) Consider a two-level atomic system with energy levels 1 and 2, which have overall lifetimes τ_1 and τ_2. In the absence of amplifier radiation, write down the rate equations for the population densities N_1 and N_2, given constant pumping rates R_1 and R_2 into each level, and a decay lifetime τ_{21} from level 2 to level 1.

   (c) Under steady-state conditions, solve for the population difference $N_0 = N_2 - N_1$. State three physical conditions regarding pumping rates and lifetimes that are required to achieve a large positive population inversion.

   (d) Now introduce amplifier radiation characterised by a transition probability density W_i for both stimulated emission and absorption. Extend your rate equations to include these processes and solve for the new steady-state population difference N in terms of N_0, W_i, and the saturation time constant τ_s.

   (e) Define the term 'gain saturation' and sketch the population difference N as a function of W_i. Explain why even very strong radiation cannot convert a negative population difference into a positive one.

6. This question considers Fourier optics, spatial filtering, and super-resolution microscopy.

   (a) A thin spherical lens can be used to isolate individual plane waves and compute a Fourier transform. State the relationship between the complex amplitude $f(x, y)$ at the front focal plane of a lens of focal length f, and the complex amplitude $g(x, y)$ at the back focal plane.

   (b) A 4-f imaging system is constructed using two identical thin lenses of focal length f. Sketch the setup, carefully labelling the object plane, the Fourier plane, and the image plane. Explain what happens to the complex amplitude of the object at the Fourier plane and the image plane.

   (c) A mask acting as a spatial low-pass filter is placed in the Fourier plane of the 4-f system. Describe the effect this mask will have on the resulting image of an object that contains fine, sharp details.

   (d) State the general equation for the point spread function of a confocal microscope and explain how it differs from a conventional widefield microscope.

   (e) Briefly outline the principle of Stimulated Emission Depletion (STED) microscopy. Explain how the use of a 'doughnut' beam allows STED to overcome the diffraction limit and achieve super-resolution imaging.

END OF PAPER